CHAPTER 12

Separations and Purifications

Independent exercise solving

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

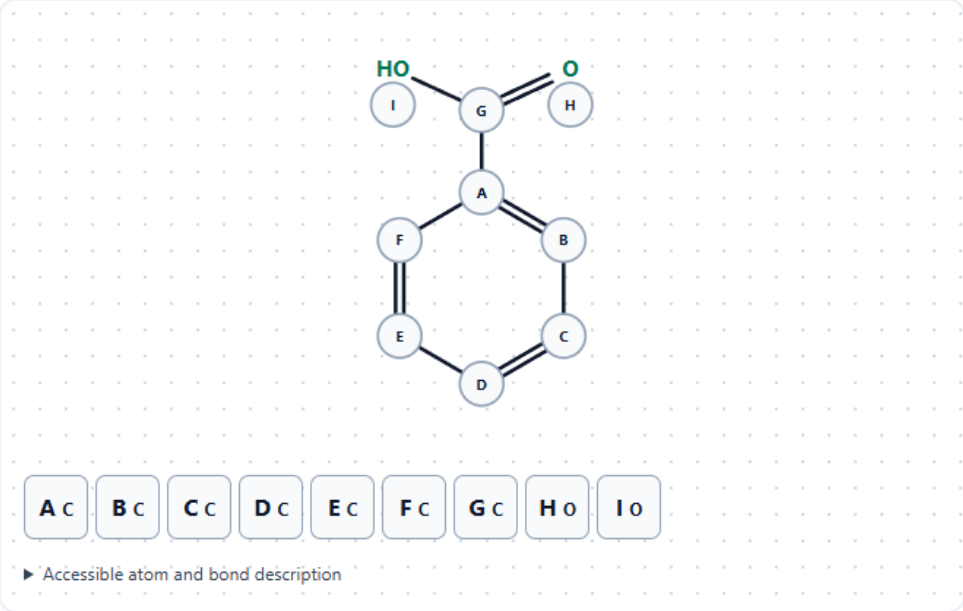
Source snapshot SHA-256: 56c72c607def508feeb45925880fcb7fc6f6dc6f8f2f40dd323eedc800e19b18

BLIND PASS: answer keys, author molecule names/formulas, hints, explanations and answer overlays are withheld. Solve every numbered stage independently. Report the stage ID, answer, derivation, and uncertainty. Question wording and displayed diagram captions are retained because they are visible task data.

Do not open the full review packet in this chat before saving your independent attempt. The full packet contains keys and lessons that would contaminate a blind attempt.

acid-extraction | Independent exercise

Screenshot: acid-extraction-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=160.00, y=95.00; charge=0; explicit H override=none

1 (B): C; x=220.00, y=130.00; charge=0; explicit H override=none

2 (C): C; x=220.00, y=200.00; charge=0; explicit H override=none

3 (D): C; x=160.00, y=235.00; charge=0; explicit H override=none

4 (E): C; x=100.00, y=200.00; charge=0; explicit H override=none

5 (F): C; x=100.00, y=130.00; charge=0; explicit H override=none

6 (G): C; x=160.00, y=35.00; charge=0; explicit H override=none

7 (H): O; x=225.00, y=5.00; charge=0; explicit H override=none

8 (I): O; x=95.00, y=5.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2], [3, 4, 1], [4, 5, 2], [5, 0, 1], [0, 6, 1], [6, 7, 2], [6, 8, 1]]

acid-extraction:stage-1 | Select the complete carboxyl group.

Answer type: atoms

acid-extraction:stage-2 | To favor transfer of this acid into an aqueous phase, choose…

Answer type: choice

Option 0 (A): A pH well above its pKa

Option 1 (B): A pH far below its pKa in every case

acid-extraction:stage-3 | To collect recrystallized solid from cold mother liquor, use…

Answer type: choice

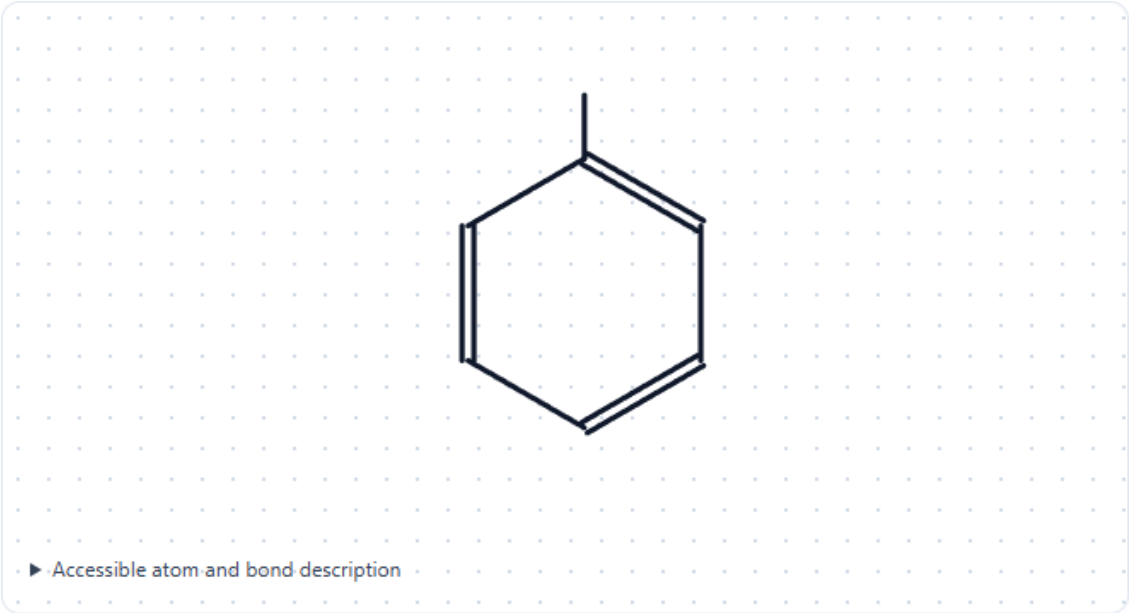
Option 0 (A): Vacuum filtration

Option 1 (B): Hot gravity filtration to keep all product dissolved

Option 2 (C): Simple distillation of the crystals

choose-distillation | Independent exercise

Screenshot: choose-distillation-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=170.00, y=37.00; charge=0; explicit H override=none

1 (B): C; x=228.89, y=71.00; charge=0; explicit H override=none

2 (C): C; x=228.89, y=139.00; charge=0; explicit H override=none

3 (D): C; x=170.00, y=173.00; charge=0; explicit H override=none

4 (E): C; x=111.11, y=139.00; charge=0; explicit H override=none

5 (F): C; x=111.11, y=71.00; charge=0; explicit H override=none

6 (G): C; x=170.00, y=5.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2], [3, 4, 1], [4, 5, 2], [5, 0, 1], [0, 6, 1]]

choose-distillation:stage-1 | A close-boiling liquid mixture is generally better separated with…

Answer type: choice

Option 0 (A): Fractional distillation

Option 1 (B): Simple filtration

choose-distillation:stage-2 | Reducing external pressure generally makes the boiling temperature…

Answer type: choice

Option 0 (A): Lower

Option 1 (B): Higher

Option 2 (C): Exactly unchanged

choose-distillation:stage-3 | Which can limit ordinary fractional distillation?

Answer type: choice

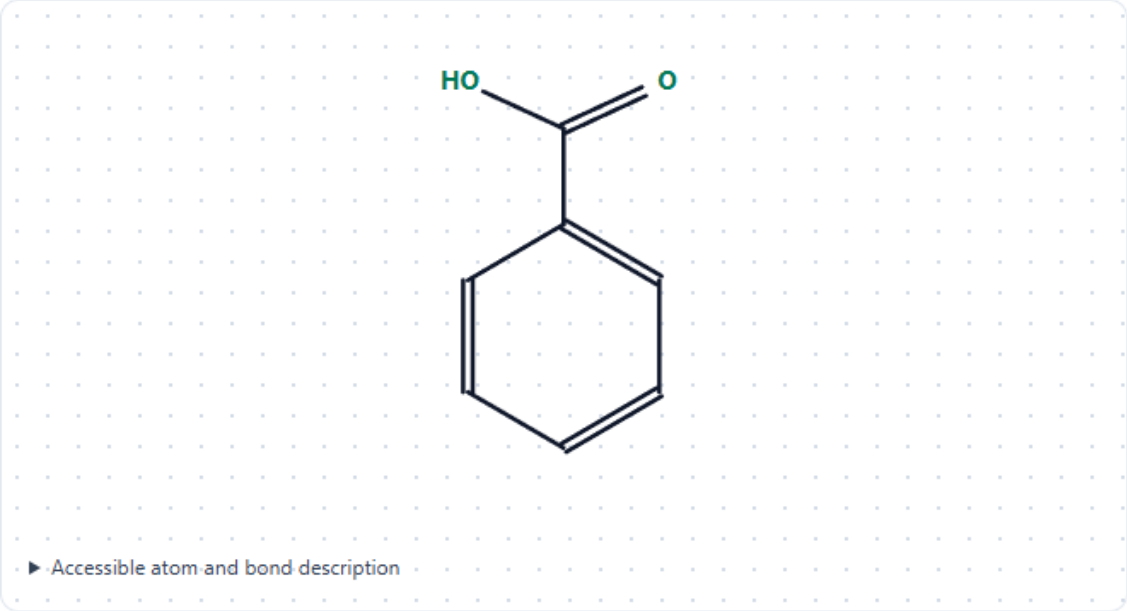
Option 0 (A): An azeotrope

Option 1 (B): The presence of any carbon atom

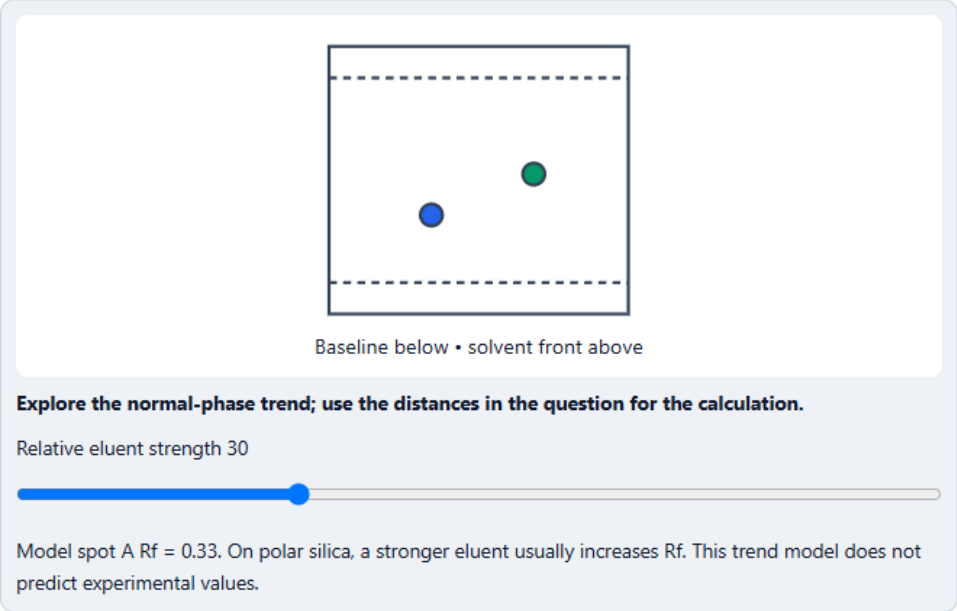
Option 2 (C): A colored liquid by itself

rf-calculation | Independent exercise

Screenshot: rf-calculation-stage-1-molecule



Screenshot: rf-calculation-diagram



{"diagramSpecification": {"type": "tlc", "caption": "Explore the normal-phase trend; use the distances in the question for the calculation."}}

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=160.00, y=95.00; charge=0; explicit H override=none

1 (B): C; x=220.00, y=130.00; charge=0; explicit H override=none

2 (C): C; x=220.00, y=200.00; charge=0; explicit H override=none

3 (D): C; x=160.00, y=235.00; charge=0; explicit H override=none

4 (E): C; x=100.00, y=200.00; charge=0; explicit H override=none

5 (F): C; x=100.00, y=130.00; charge=0; explicit H override=none

6 (G): C; x=160.00, y=35.00; charge=0; explicit H override=none

7 (H): O; x=225.00, y=5.00; charge=0; explicit H override=none

8 (I): O; x=95.00, y=5.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2], [3, 4, 1], [4, 5, 2], [5, 0, 1], [0, 6, 1], [6, 7, 2], [6, 8, 1]]

rf-calculation:stage-1 | A spot center travels 1.8 cm and the solvent front 6.0 cm. What is Rf?

Answer type: number | unit: dimensionless

rf-calculation:stage-2 | On normal-phase silica, a stronger eluent usually makes Rf…

Answer type: choice

Option 0 (A): Increase

Option 1 (B): Decrease in every case

rf-calculation:stage-3 | Two compounds with the same Rf on one plate are…

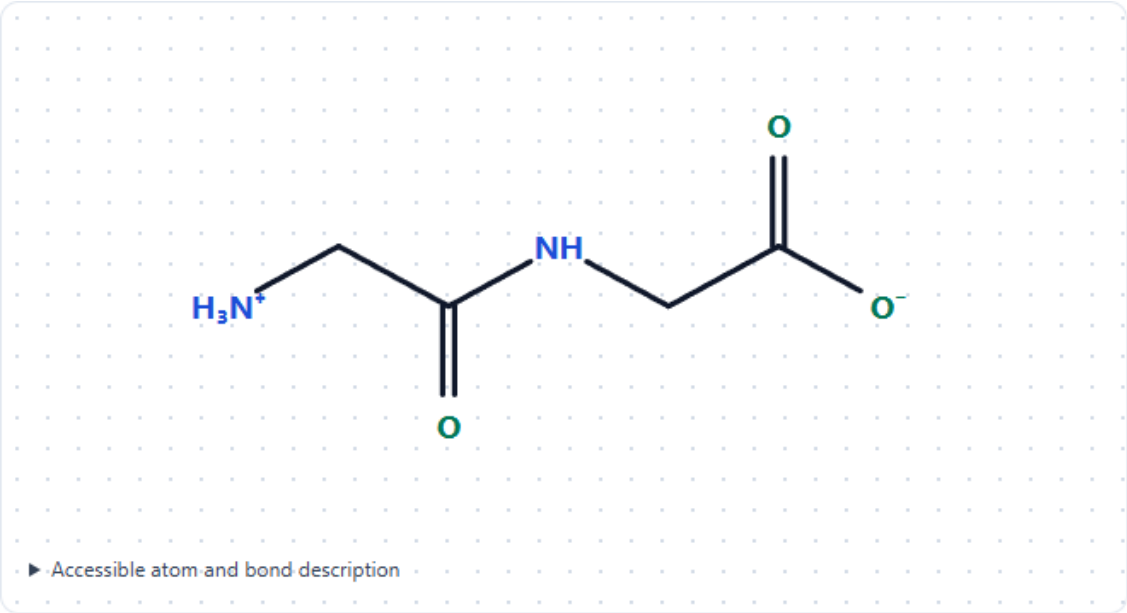
Answer type: choice

Option 0 (A): Not proven identical

Option 1 (B): Necessarily identical

chromatography-match | Independent exercise

Screenshot: chromatography-match-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): N; x=35.00, y=110.00; charge=1; explicit H override=none

1 (B): C; x=90.00, y=80.00; charge=0; explicit H override=none

2 (C): C; x=145.00, y=110.00; charge=0; explicit H override=none

3 (D): O; x=145.00, y=170.00; charge=0; explicit H override=none

4 (E): N; x=200.00, y=80.00; charge=0; explicit H override=none

5 (F): C; x=255.00, y=110.00; charge=0; explicit H override=none

6 (G): C; x=310.00, y=80.00; charge=0; explicit H override=none

7 (H): O; x=310.00, y=20.00; charge=0; explicit H override=none

8 (I): O; x=365.00, y=110.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [2, 4, 1], [4, 5, 1], [5, 6, 1], [6, 7, 2], [6, 8, 1]]

chromatography-match:stage-1 | A protein at pH 7 with pI 5 tends to have what net charge?

Answer type: choice

Option 0 (A): Negative

Option 1 (B): Positive

chromatography-match:stage-2 | Which resin would electrostatically bind that protein?

Answer type: choice

Option 0 (A): Positively charged anion exchanger

Option 1 (B): Negatively charged cation exchanger

chromatography-match:stage-3 | In ideal size exclusion, larger molecules within the useful separation range usually elute...

Answer type: choice

Option 0 (A): Earlier

Option 1 (B): Later